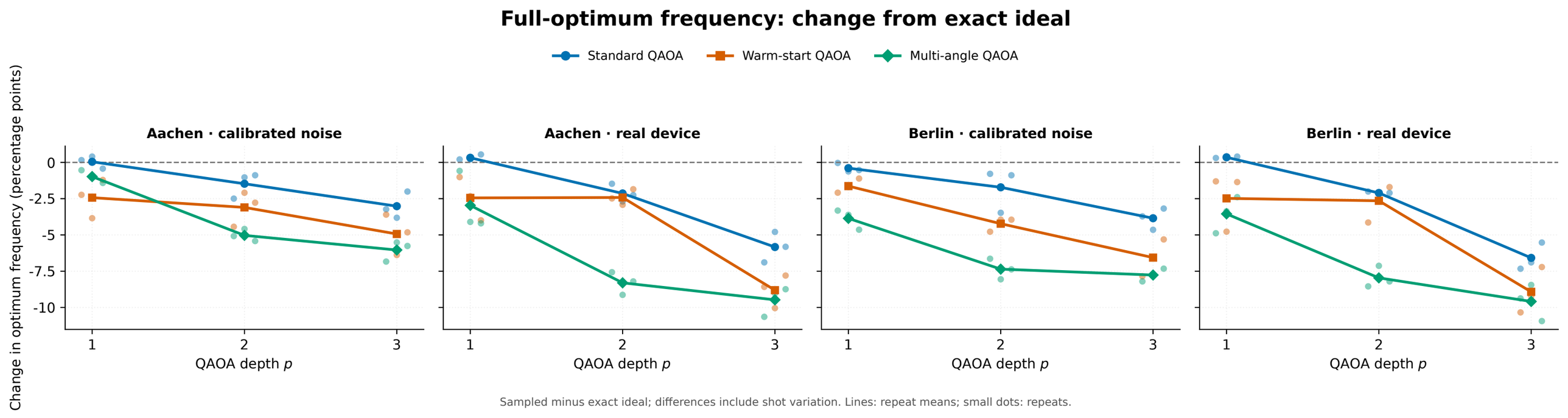

Full-optimum frequency: change from exact ideal
Standard QAOA
Warm-start QAOA
Multi-angle QAOA
Aachen · calibrated noise
Aachen · real device
Berlin · calibrated noise
Berlin · real device
Change in optimum frequency (percentage points)
0
-2.5
-5
-7.5
-10
1
2
3
QAOA depth p
Sampled minus exact ideal; differences include shot variation. Lines: repeat means; small dots: repeats.